M13D — Mock exam 3

Instruments · Rotary Wing Aerodynamics · Autoflight · Equipment and Furnishings

40 questions · 20 Instruments · 13 Autoflight · 5 Rotary Wing Aerodynamics · 2 Equipment and Furnishings

Level 3: 22 · Level 2: 13 · Level 1: 5. Three options, one correct.

Every question sits on a fact highlighted or underlined in the manual. No question in this paper appears in any of the other four.

The answer key follows the questions, with the explanation and the manual PDF page for each one.

INSTRUMENTS

1. Why are static ports placed in areas where turbulence and ram air distortion are low
   - A. Because turbulence and ram air distortion overheat the static port zones
   - B. Because turbulence and ram air distortion block the static pressure hoses
   - C. Because turbulence and ram air distortion falsify the static pressure reading

2. A Prandtl tube delivers Ptot and Ps to a computer. Which quantity does the computer obtain from them, and what is it needed for?
   - A. The ram pressure, needed for the speed indication
   - B. The total air temperature, needed for the speed indication
   - C. The static source error, needed for the altitude indication

3. The limits for damage to the skin at the static ports are found
   - A. In the ATA chapter for the instruments or the IPC
   - B. In the ATA chapter for the aircraft structure or the SRM
   - C. In the ATA chapter for air conditioning or the wiring manual

4. Why can water be present in the pressure pipes
   - A. They are not drained on a regular basis
   - B. They are capped for protection during testing
   - C. Their test fittings are used to simulate altitude and speed

5. TAS is obtained from IAS by correcting for instrument error, static source error, dynamic pressure error and
   - A. Magnetic variation
   - B. Aircraft weight
   - C. Air density

6. An aircraft with an IVSI enters and leaves a turn with a high bank angle. What should be expected?
   - A. A correct vertical speed indication, because the piston compensates for bank angle
   - B. An inaccurate vertical speed indication, because any vertical acceleration affects the piston
   - C. A total loss of indication, because the piston is locked by the acceleration

7. A servo corrected altimeter reads a long way from the altitude sent by the air data computer. Why does the correction not bring it into agreement?
   - A. Correction is inhibited while the internal vibrator is running
   - B. Correction only works above the altitude set on the baro scale
   - C. Correction is only possible within a few hundred feet

8. An air data computer stops working altogether. What is seen on the Mach/airspeed indicator?
   - A. A red flag for the Mach value only, because CAS is taken from a capsule
   - B. A red flag for each value, because the valid discretes are no longer transmitted
   - C. No flag at all, because the indicator holds the last values received

9. An aeroplane gives no noticeable initial buffet as its speed decays towards the point of lift loss. What must be installed?
   - A. A stall warning system
   - B. An altitude alert system
   - C. An angle of attack indicator

10. The altitude readout box on the PFD changes from white to yellow. Which mode has the system entered?
   - A. Deviation mode
   - B. Capture mode
   - C. Approach mode

11. Health and usage monitoring systems are a common part of the flight avionics of
   - A. a modern large aeroplane
   - B. a modern helicopter
   - C. a modern business jet

12. A free gyroscope is left standing at the equator. Which of the two apparent errors is at its greatest there?
   - A. transport wander
   - B. apparent drift
   - C. apparent topple

13. On which law of magnetism is the function of the magnetic compass based
   - A. Unlike poles attract and like poles repel
   - B. All poles attract when they are close enough
   - C. Like poles attract and unlike poles repel

14. Which colour indicates the normal operation of a temporarily used system
   - A. Blue
   - B. Green
   - C. Amber

15. The aural message SINK RATE has sounded but PULL UP has not. Which boundary has the aircraft penetrated?
   - A. The outer alert boundary
   - B. The upper alert boundary
   - C. The inner alert boundary

16. The flaps are at the landing position and the aircraft is within two dots of the localizer on an ILS approach, so which submode boundary applies?
   - A. Mode 2A
   - B. Mode 4B
   - C. Mode 2B

17. The aircraft drifts 1.5 dots below the beam centre at 800 feet radio altitude. Which alert, if any, is triggered?
   - A. The hard alert
   - B. No alert at all
   - C. The soft alert

18. An aural message is heard and exactly seven seconds later the same words are heard again. Which envelope is still penetrated?
   - A. The warning envelope
   - B. The caution envelope
   - C. The clearance floor envelope

19. The fuel level in a tank changes. What does the multi-tubular surface condenser in the tank do?

A. It changes its capacity
B. It changes its excitation frequency
C. It changes its resistance

20. The output frequency of a densitometer is read. What is that frequency proportional to?

A. The fuel height
B. The fuel temperature
C. The fuel density

AUTOFLIGHT

21. A controlled system is acted on by disturbances whose effect cannot be predicted. Why is open loop control the wrong choice here?

A. It can only correct for what was allowed for in advance
B. It needs a measurement unit that unpredictable disturbances overload
C. It feeds the disturbance back and so amplifies the error further

22. In a P-regulator the change in the output value is always proportional to

A. the magnitude of the input value
B. the change in the input value
C. the rate of change of the input value

23. As the aircraft accelerates past a certain airspeed the maximum rudder deflection available is reduced. What does this limiting achieve?

A. It keeps the yaw damper inside its authority
B. It keeps the loads within the limits of structural strength
C. It keeps the surface within its maximum deflection angle

24. The autopilot is engaged while the aircraft is banked three degrees. What does the autopilot do?

A. It holds the three degrees of bank and lets the heading run
B. It rolls back to wings level and holds the heading of that moment
C. It rolls back to wings level and holds the last selected heading

25. An aircraft climbs through the changeover altitude with a speed target that was dialled in by hand. What becomes of the number in the window?

A. It is converted to the equivalent Mach number and displayed there
B. It is converted to the equivalent true airspeed and displayed there
C. It is replaced by dashes until the pilot enters a Mach number

26. In the vertical navigation modes, which channel performs the vertical guidance?

A. The pitch channel
B. The autothrust channel
C. The roll channel

27. Beam convergence has been left uncompensated on an ILS approach — how does the glide path controller behave at long range and close in?

A. Sluggish far out and unable to hold the beam near the runway
B. Prone to oscillation far out and sluggish near the runway
C. Sluggish far out and prone to oscillation near the runway

28. The preselected course is the reference for the align phase — what must the pilot check before the alignment?

- **A.** That it agrees with the present heading
- **B.** That it agrees with the runway heading
- **C.** That it agrees with the localizer frequency

29. A crosswind automatic landing is flown — what does the flight mode annunciator show as the yaw align phase begins?

- **A.** ROLLOUT appears as the lateral mode
- **B.** ALIGN appears as the lateral mode
- **C.** Nothing at all is annunciated for it

30. Which channel and which surface does the align phase require

- **A.** A roll channel acting on the ailerons
- **B.** A yaw channel acting on the nose wheel
- **C.** A yaw channel acting on the rudder

31. A large thrust increase at the start of a go around produces a strong nose up moment — which limits must the go around controller watch?

- **A.** The pitch attitude or angle of attack limits
- **B.** The bank angle or sideslip limits
- **C.** The flap or landing gear speed limits

32. An engine fails while the aircraft is accelerating and is still below V1. What must be done?

- **A.** The take off has to be discontinued
- **B.** The aircraft has to be rotated at once
- **C.** The take off has to be continued

33. The pitch channel is given the task of holding a certain altitude. Which system then controls the airspeed?

- **A.** The pitch channel, which keeps its priority
- **B.** The auto-throttle or auto-thrust system
- **C.** The flight management system computer

ROTARY WING AERODYNAMICS

34. Which cockpit input changes the pitch angle of all main rotor blades

- **A.** The tail rotor pedals
- **B.** The cyclic pitch stick
- **C.** The collective pitch control

35. A pilot raises the collective lever to climb and the torque effect on the fuselage changes. Which control must the pilot then use to vary the thrust that compensates it?

- **A.** The tail rotor pedals
- **B.** The collective lever
- **C.** The cyclic pitch stick

36. A helicopter is hovering with no wind at all. Why is there no dissymmetry of lift in this condition?

- **A.** There is no pitch angle on either blade, so both blades have the same parameters
- **B.** There is no airstream due to forward speed, so both blades have the same parameters
- **C.** There is no rotational speed at the blade tips, so both blades have the same parameters

37. Why does upward flapping reduce the angle of attack of the advancing blade?

- **A.** Moving up reduces the blade pitch angle
- **B.** Moving up adds a downward relative wind component
- **C.** Moving up adds an upward relative wind component

38. Once the engine has failed, what does the collective pitch lever control

- **A.** The tail rotor thrust that keeps the heading in the descent
- **B.** The engine power that keeps the rotor turning in the descent
- **C.** The force that keeps the rotor turning in the descent

EQUIPMENT AND FURNISHINGS

39. Two operators want the cabin of the same aeroplane arranged to suit their own service. What does the CIDS design philosophy give them?

- **A.** A fixed configuration set once by the airframe manufacturer
- **B.** Flexibility gained by exchanging the cabin wiring looms
- **C.** Flexibility without complex and time-consuming hardware changes

40. For which aircraft is the NICE cabin management system offered

- **A.** Long range freighter aircraft
- **B.** VIP and government aircraft
- **C.** Regional commuter aircraft

Answer key

Answer, explanation and the manual page each question comes from

1. **C** A port in a quiet zone senses the true static pressure, while turbulence and ram air distortion at the port falsify that reading. (L3, PDF p. 407)
2. **A** The differential of the two delivered pressures is the ram pressure q, which is what the speed indication requires. (L3, PDF p. 415)
3. **B** Skin damage is a structural item, so the structure chapter or the SRM holds those limits. (L2, PDF p. 425)
4. **A** The pipes are not drained routinely, so water can collect in them. (L2, PDF p. 443)
5. **C** Air density is the fourth correction listed for deriving TAS from IAS. (L2, PDF p. 471)
6. **B** Any vertical acceleration acts on the piston, so turn entry and exit give a wrong reading. (L2, PDF p. 483)
7. **C** Continuous correction requires the error voltage to be about zero, which limits it to a few hundred feet. (L3, PDF p. 501)
8. **B** Loss of the valid discrete for a value raises the red flag for that value. (L3, PDF p. 507)
9. **A** A stall warning system must be installed when the approaching stall speed is not immediately noticeable through initial buffet. (L2, PDF p. 555)
10. **A** A white to yellow change of the altitude readout box marks the deviation mode. (L3, PDF p. 567)
11. **B** Health and usage monitoring systems are a common component in the flight avionics of modern helicopters. (L1, PDF p. 615)
12. **C** Apparent topple is greatest at the equator. (L3, PDF p. 627)
13. **A** The main law of magnetism quoted is that unlike poles attract and like poles repel. (L1, PDF p. 737)
14. **A** Blue indicates the normal operation of a temporarily used system. (L1, PDF p. 860)
15. **A** SINK RATE is the outer boundary message; PULL UP would mean the inner boundary had been crossed as well. (L3, PDF p. 909)
16. **C** That configuration is one of those listed for mode 2B. (L3, PDF p. 911)
17. **C** The aircraft is beyond 1.3 dots below the beam centre, so the soft alert comes; the hard alert needs below 300 feet radio altitude and more than 2 dots. (L3, PDF p. 917)
18. **B** A seven second repeat is the caution level; the warning level repeats continuously. (L3, PDF p. 927)
19. **A** The condenser changes its capacity depending on the level of fuel in the tank. (L2, PDF p. 979)
20. **C** The frequency of the vibrations is proportional to the fuel density. (L2, PDF p. 993)
21. **A** Without feedback the correction must be worked out beforehand, which an unpredictable disturbance makes impossible. (L3, PDF p. 68)
22. **B** A proportional regulator ties the change of its output directly to the change of its input. (L1, PDF p. 72)
23. **B** Because effectiveness grows with dynamic pressure, deflection is limited at speed to hold loads within structural limits. (L3, PDF p. 100)
24. **B** Under five degrees the bank angle criterion selects heading hold on the heading present at engagement. (L3, PDF p. 118)
25. **A** At the transition the value is converted to the equivalent Mach number, and a value entered at the panel is shown there. (L3, PDF p. 150)
26. **A** Vertical guidance is a pitch task and thrust control is left to the autothrust. (L2, PDF p. 176)
27. **C** Low gain far out makes it sluggish, and rising gain close in makes it oscillate. (L3, PDF p. 206)
28. **B** The align phase is only as good as the preselected course, so it must match the runway heading. (L2, PDF p. 218)
29. **C** The align phase gets no annunciation at all. (L2, PDF p. 218)
30. **C** The decrab is flown by the yaw channel through the rudder. (L2, PDF p. 218)
31. **A** The required thrust increase causes a large positive pitch moment, so the pitch attitude and angle of attack limits have to be observed. (L3, PDF p. 226)

32. **A** Before V1 an engine failure obliges the crew to stop the take off. (L3, PDF p. 238)
33. **B** The pitch channel delegates speed control once it is busy with another task. (L3, PDF p. 252)
34. **C** The collective pitch control changes the pitch angle. (L1, PDF p. 18)
35. **A** Torque is compensated by the tail rotor, whose thrust the pilot varies with the pedals. (L3, PDF p. 42)
36. **B** Without the airstream caused by the speed of flight there is nothing to add to one blade and subtract from the other. (L2, PDF p. 46)
37. **B** Relative wind is opposite to blade motion, so an upward-moving blade meets air from above, reducing the angle of attack. (L3, PDF p. 12)
38. **C** With no engine the collective no longer sets engine power; it controls the force that sustains rotor RPM. (L3, PDF p. 60)
39. **C** The CIDS is designed for high flexibility so that complex and time expensive hardware changes are not needed. (L3, PDF p. 334)
40. **B** The system is offered for VIP and government aircraft. (L2, PDF p. 338)